

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 12: Surface Areas and Volumes

CHAPTER 12: SURFACE AREAS AND VOLUMES

A simplified, detailed guide in Hinglish

For students who missed Class 7-8-9

Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 SURFACE AREA VS VOLUME

SURFACE AREA (SA): solid ke bahar ki total area
(paint karne layak). Unit = cm^2 / m^2 .

VOLUME: solid ke andar ki jagah (paani bharne layak).
Unit = cm^3 / m^3 .

Do tarah ki surface area:

- CSA (Curved/Lateral): sirf curved/side part
(top-bottom chhod ke)
- TSA (Total): sab kuch (curved + flat faces)

1.2 pi REMINDER

$\pi = 22/7$ (radius 7 ka multiple) ya 3.14

SECTION 2: BASIC SOLIDS KE FORMULAS

2.1 CUBE (side = a)

$$\text{TSA} = 6a^2$$

$$\text{Volume} = a^3$$

2.2 CUBOID (l, b, h)

$$\text{TSA} = 2(lb + bh + hl)$$

$$\text{Volume} = l \times b \times h$$

2.3 CYLINDER (radius r, height h)

$$\text{CSA} = 2 \pi r h$$

$$\text{TSA} = 2 \pi r (r + h)$$

$$\text{Volume} = \pi r^2 h$$

2.4 CONE (radius r, height h, slant l)

$$\text{slant height } l = \sqrt{r^2 + h^2}$$

$$\text{CSA} = \pi r l$$

$$\text{TSA} = \pi r (l + r)$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

2.5 SPHERE (radius r)

$$\text{Surface Area} = 4 \pi r^2$$

$$\text{Volume} = \frac{4}{3} \pi r^3$$

2.6 HEMISPHERE (radius r) - aadha sphere

$$\text{CSA} = 2 \pi r^2$$

$$\text{TSA} = 3 \pi r^2 \text{ (curved + flat circle)}$$

$$\text{Volume} = \frac{2}{3} \pi r^3$$

SECTION 3: EXAMPLES (SINGLE SOLIDS)

EXAMPLE 1: Cylinder $r=7$ cm, $h=10$ cm. Volume & CSA.

$$(\pi = 22/7)$$

$$\text{Volume} = \pi r^2 h = (22/7) \times 7 \times 7 \times 10$$

$$= 22 \times 7 \times 10 = 1540 \text{ cm}^3$$

$$\text{CSA} = 2 \pi r h = 2 \times (22/7) \times 7 \times 10 = 440 \text{ cm}^2$$

EXAMPLE 2: Cone $r=6$ cm, $h=8$ cm. Slant & volume.

$$l = \sqrt{r^2 + h^2} = \sqrt{36 + 64} = \sqrt{100} = 10 \text{ cm}$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times 3.14 \times 36 \times 8 = 301.44 \text{ cm}^3$$

EXAMPLE 3: Sphere $r=7$ cm. Surface area. ($\pi = 22/7$)

$$\text{SA} = 4 \pi r^2 = 4 \times (22/7) \times 7 \times 7$$

$$= 4 \times 22 \times 7 = 616 \text{ cm}^2$$

SECTION 4: COMBINATION OF SOLIDS

Class 10 mein zyadatar solids JODKE banaye jaate hain -
jaise ice-cream (cone + hemisphere), capsule (cylinder
+ 2 hemispheres), etc.

4.1 GOLDEN RULE

Jab solids jodte hain:

- Surface area: sirf EXPOSED (dikhne wale) surfaces
ADD karo. (chhupa hua surface mat ginna)
- Volume: hamesha simply ADD karo.

4.2 EXAMPLE 4: ICE-CREAM (cone + hemisphere on top)

Cone ($r=3.5$, $h=12$) ke upar hemisphere ($r=3.5$).
Total volume. ($\pi = 22/7$)

$$\begin{aligned}\text{Cone vol} &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 12 \\ &= 154 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Hemisphere vol} &= \frac{2}{3} \pi r^3 \\ &= \frac{2}{3} \times \frac{22}{7} \times 42.875 \\ &= 89.83 \text{ cm}^3 \text{ (approx)}\end{aligned}$$

$$\text{Total} = 154 + 89.83 = 243.83 \text{ cm}^3 \text{ (approx)}$$

4.3 SURFACE AREA LOGIC (CAPSULE)

Capsule = cylinder + 2 hemispheres (ends).

$$\begin{aligned}\text{Total SA} &= (\text{CSA cylinder}) + 2 \times (\text{CSA hemisphere}) \\ &= 2 \pi r h + 2 \times (2 \pi r^2)\end{aligned}$$

Cylinder ke flat ends count NAHI (hemisphere se dhake).

SECTION 5: FRUSTUM OF A CONE (ADVANCED)

FRUSTUM = cone ka upar wala hissa kaat do, neeche
jo bachta hai (jaise bucket, glass).

Radii R (bada, neeche), r (chhota, upar), height h :

$$\text{slant } l = \sqrt{h^2 + (R - r)^2}$$

$$\begin{aligned} \text{CSA} &= \pi (R + r) l \\ \text{TSA} &= \pi (R + r) l + \pi R^2 + \pi r^2 \\ \text{Volume} &= \frac{1}{3} \pi h (R^2 + r^2 + Rr) \end{aligned}$$

Q AND A TIME

Q1. Cube ka side 5 cm. TSA aur volume nikaal.

Q2. Cylinder $r=7$ cm, $h=20$ cm. Volume. ($\pi = 22/7$)

Q3. Cone $r=9$ cm, $h=12$ cm. Slant height l nikaal.

Q4. Sphere $r=3$ cm. Volume nikaal. ($\pi = 3.14$)

Q5. Capsule (cylinder + 2 hemispheres) ka surface area kaise nikaalte hain? Logic bata.

Q6. Hemisphere ka TSA aur CSA mein kya difference? (formula bata)

SUMMARY (FORMULA CHEAT SHEET)

Solid	CSA/LSA	TSA/SA	Volume
Cube	$4a^2$	$6a^2$	a^3
Cuboid	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cylinder	$2\pi r h$	$2\pi r(r+h)$	$\pi r^2 h$
Cone	$\pi r l$	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Sphere	-	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

Combination rule:

SA -> only EXPOSED surfaces add karo

Volume -> simply add

Cone slant: $l = \sqrt{r^2 + h^2}$

TIPS FOR EXAM

1. Saare formulas ki cheat sheet HARD yaad kar -
ye chapter pura formula-based hai.
2. Combination problems mein diagram banao aur decide
karo konsa surface dikhta hai (exposed) - wahi add.
3. Volume hamesha simply add hota hai - SA mein
dhyaan rakhna padta hai.
4. Units check kar: SA $\rightarrow$ cm^2 , Volume $\rightarrow$ cm^3 .
5. Cone mein l (slant) aur h (height) alag hote hain -
 $l = \sqrt{r^2 + h^2}$. Confuse mat ho.